HW # 6

1. (a) $x(n) = a^n[u(n) - u(n-n_0)]$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} a^n[u(n)-u(n-n_0)]e^{-j\Omega n} = \sum_{n=0}^{n_0-1} a^n e^{-j\Omega n} = \sum_{n=0}^{n_0-1}(ae^{-j\Omega})^n$$

$$= \frac{1-(ae^{-j\Omega})^{n_0}}{1-ae^{-j\Omega}}$$

(b) $x(n) = n\cdot\left(\frac{1}{3}\right)^{|n|}$

let $x_1(n) = \left(\frac{1}{3}\right)^{|n|}$ $\quad X_1(\Omega) = \sum_{n=-\infty}^{\infty} x_1(n)e^{-j\Omega n} = \sum_{n=-\infty}^{-1}\left(\frac{1}{3}\right)^{-n}e^{-j\Omega n} + \sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n e^{-j\Omega n}$

let $m=-n$

$$= \sum_{m=1}^{\infty}\left(\frac{1}{3}\right)^m e^{j\Omega m} + \sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n e^{-j\Omega n}$$

$$= \sum_{m=1}^{\infty}\left(\frac{1}{3}e^{j\Omega}\right)^m + \sum_{n=0}^{\infty}\left(\frac{1}{3}e^{-j\Omega}\right)^n$$

$$= \frac{\frac{1}{3}e^{j\Omega}}{1-\frac{1}{3}e^{j\Omega}} + \frac{1}{1-\frac{1}{3}e^{-j\Omega}}$$

$$= \frac{\frac{1}{3}e^{j\Omega}(1-\frac{1}{3}e^{-j\Omega}) + (1-\frac{1}{3}e^{j\Omega})}{(1-\frac{1}{3}e^{j\Omega})(1-\frac{1}{3}e^{-j\Omega})}$$

$$= \frac{\frac{1}{3}e^{j\Omega} - \frac{1}{9} + 1 - \frac{1}{3}e^{j\Omega}}{1-\frac{1}{3}(e^{j\Omega}+e^{-j\Omega})+\frac{1}{9}}$$

$$= \frac{\frac{8}{9}}{\frac{10}{9}-\frac{2}{3}\cos\Omega} = \frac{8}{10-6\cos\Omega} = \frac{4}{5-3\cos\Omega}$$

so that $X(\Omega) = j\cdot\frac{dX_1(\Omega)}{d\Omega} = j\cdot\frac{d}{d\Omega}\left(\frac{4}{5-3\cos\Omega}\right)$

$$= j\cdot\frac{-4\cdot(5-3\cos\Omega)'}{(5-3\cos\Omega)^2}$$

$$= j\cdot\frac{-4\cdot(-3)\cdot(-\sin\Omega)}{(5-3\cos\Omega)^2}$$

$$= \frac{-12j\sin\Omega}{(5-3\cos\Omega)^2}$$

ce). $x(n) = a^n\cos(\Omega_0 n)u(n)$ $\quad |a|<1$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} a^n\cos(\Omega_0 n)\cdot u(n)e^{-j\Omega n} = \sum_{n=0}^{\infty}\frac{1}{2}\left[a^n e^{j\Omega_0 n} + a^n e^{-j\Omega_0 n}\right]e^{-j\Omega n}$$

$$= \frac{1}{2}\sum_{n=0}^{\infty}\left[a^n e^{j(\Omega_0-\Omega)n} + a^n e^{-j(-\Omega_0+\Omega)n}\right]$$

$$= \frac{1}{2}\cdot\left[\frac{1}{1-ae^{j(\Omega_0-\Omega)}} + \frac{1}{1-ae^{-j(\Omega_0+\Omega)}}\right]$$

$$= \frac{1-ae^{-j\Omega}\cos\Omega_0}{1-2ae^{-j\Omega}\cos\Omega_0 + a^2e^{-j2\Omega}}$$

(2)

(a). $x(n) = (n-1)a^n u(n) \quad |a|<1$

$x(n) = na^n u(n) - a^n u(n)$ since, $x_2(n) \longleftrightarrow X_2(\Omega)$, $a^n u(n) \longleftrightarrow \frac{e^{j\Omega}}{e^{j\Omega}-a}$

$$\frac{dX_2(\Omega)}{d\Omega} = \frac{je^{j\Omega}(e^{j\Omega}-a) - e^{j\Omega}(je^{j\Omega})}{(e^{j\Omega}-a)^2} = \frac{je^{j2\Omega} - aje^{j\Omega} - je^{j2\Omega}}{(e^{j\Omega}-a)^2}$$

$$= -j\frac{ae^{j\Omega}}{(e^{j\Omega}-a)^2}$$

$x_1(n) \longleftrightarrow X_1(\Omega)$, $na^n u(n) \longleftrightarrow j\frac{dX_2(\Omega)}{d\Omega}$

$$X_1(\Omega) = j\cdot(-j)\cdot\frac{ae^{j\Omega}}{(e^{j\Omega}-a)^2} = \frac{ae^{j\Omega}}{(e^{j\Omega}-a)^2}$$

So: $X(\Omega) = X_1(\Omega) - X_2(\Omega)$

$$= \frac{ae^{j\Omega}}{(e^{j\Omega}-a)^2} - \frac{e^{j\Omega}}{(e^{j\Omega}-a)}$$

$$= \frac{ae^{j\Omega} - e^{j\Omega}(e^{j\Omega}-a)}{(e^{j\Omega}-a)^2} = \boxed{\frac{2ae^{j\Omega} - e^{j2\Omega}}{(e^{j\Omega}-a)^2}}$$

2. (a) $X(\Omega) = -j\pi\delta(\Omega - \frac{\pi}{3}) + \pi\delta(\Omega - \frac{2}{3}\pi)$

since $x(n) = \frac{1}{2\pi}\int_0^{2\pi} X(\Omega)e^{j\Omega n}d\Omega$

$$x_1(n) = \frac{1}{2\pi}\int_{\langle 2\pi\rangle}\left\{\left[-j\pi\delta(\Omega-\frac{\pi}{3})e^{j\Omega n}\right] + \left[\pi\delta(\Omega-\frac{2}{3}\pi)e^{j\Omega n}\right]\right\}d\Omega$$

$$= \frac{1}{2\pi}\int_{\langle 2\pi\rangle}\left\{\left[-j\pi\delta(\Omega-\frac{\pi}{3})\cdot e^{j\frac{\pi}{3}n}\right] + \left[\pi\delta(\Omega-\frac{2}{3}\pi)e^{j\frac{2}{3}\pi n}\right]\right]d\Omega$$

$$= \frac{1}{2\pi}\left[-j\pi e^{j\frac{\pi}{3}n} + \pi e^{j\frac{2}{3}\pi n}\right]\int_{\langle 2\pi\rangle}\left[\delta(\Omega-\frac{\pi}{3}) + \delta(\Omega-\frac{2}{3}\pi)\right]d\Omega$$

$$= \frac{1}{2\pi}\left(-j\pi e^{-j\frac{\pi}{3}n} + \pi e^{j\frac{2}{3}\pi n}\right)$$

$$\boxed{= \frac{1}{2}\left[e^{j\frac{2}{3}\pi n} - je^{-j\frac{\pi}{3}n}\right]}$$

(b) $X(\Omega) = \dfrac{4}{(\exp(-j\Omega)-2)^2}$

(3)

Let $z = e^{j\Omega}$, $X(z) = \dfrac{4}{(z^{-1}-2)^2} = \dfrac{4z^2}{(1-2z)^2} = \dfrac{z^2}{(z-\frac{1}{2})^2} = 2\dfrac{\frac{1}{2}z}{(z-\frac{1}{2})^2} \cdot z$

Since $n\alpha^n u(n) \longleftrightarrow \dfrac{\alpha z}{(z-\alpha)^2}$

$(n+1)\alpha^{n+1} u(n+1) \longleftrightarrow \dfrac{\alpha z^2}{(z-\alpha)^2}$

So that $\boxed{x(n) = 2(n+1)\left(\frac{1}{2}\right)^{n+1} u(n+1)}$

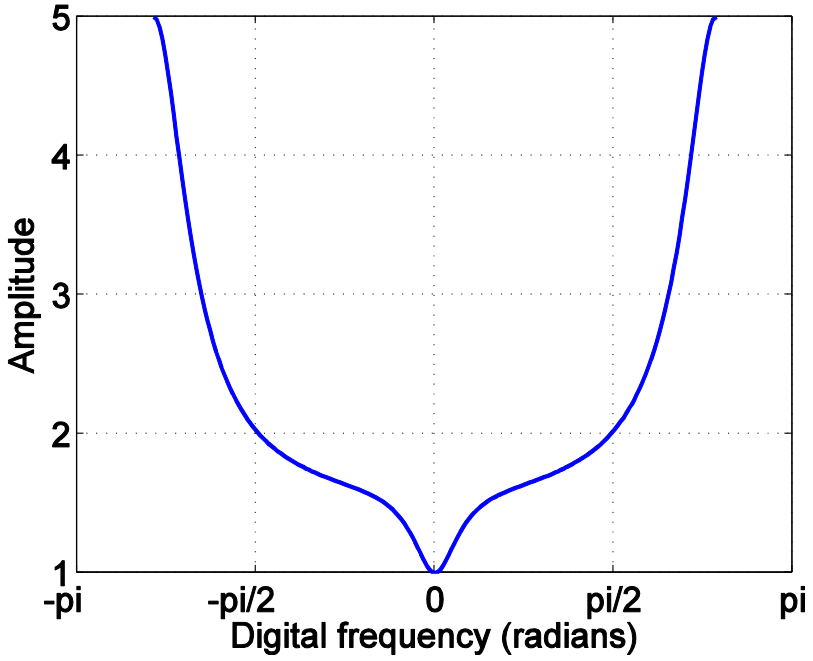
3.

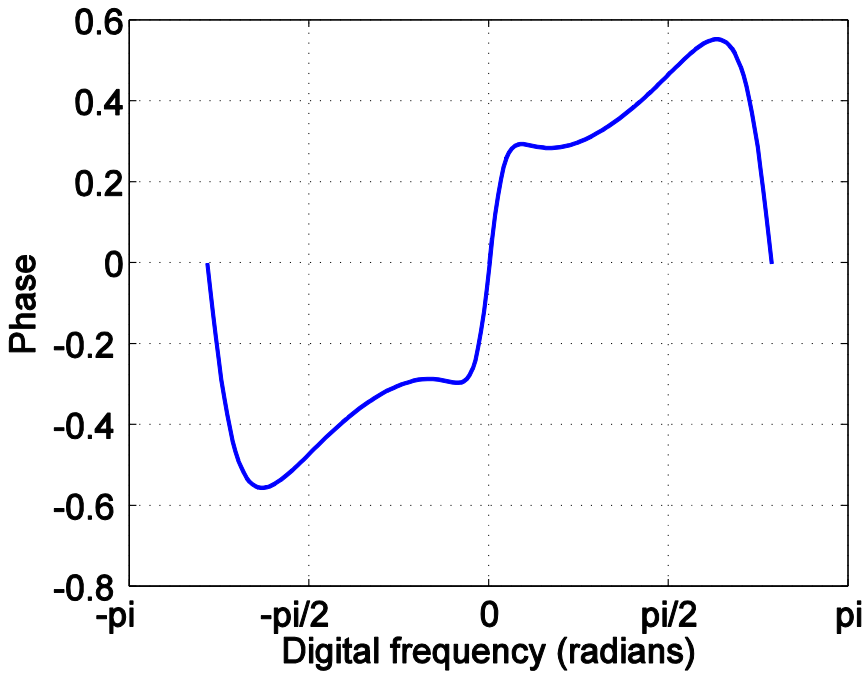
$y(n) - 0.5y(n-2) = 2x(n) - x(n-1) - 0.5x(n-2)$

$Y(\Omega) - 0.5e^{-j2\Omega}Y(\Omega) = 2X(\Omega) - e^{-j\Omega}X(\Omega) - 0.5e^{-j2\Omega}X(\Omega)$

$Y(\Omega)\left[1 - 0.5e^{-j2\Omega}\right] = X(\Omega)\left[2 - e^{-j\Omega} - 0.5e^{-j2\Omega}\right]$

$H(\Omega) = \dfrac{Y(\Omega)}{X(\Omega)} = \dfrac{2 - e^{-j\Omega} - 0.5e^{-j2\Omega}}{1 - 0.5e^{-j2\Omega}}$





Codes:

```
clear all;
clc;
step=pi/100;
omega=-pi:step:pi;
H_omega=(2-exp(-j*omega)-0.5*exp(-j*2*omega))./(1-
0.5*exp(-j*2*omega));
figure(1);
set(gca,'fontsize',16);
plot(omega,abs(H_omega),'LineWidth',1.5);
set(gca,'XTickLabel',{'-pi','-pi/2','0','pi/2','pi'})
xlabel('Digital frequency
(radians)');ylabel('Amplitude');
grid on;


figure(2);
set(gca,'fontsize',16);
plot(omega,angle(H_omega),'LineWidth',1.5);
set(gca,'XTickLabel',{'-pi','-pi/2','0','pi/2','pi'})
xlabel('Digital frequency (radians)');ylabel('Phase');
grid on;
```